# AMIRAS RABARI

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*Aspiring engineer curious about how things work, exploring systems, software, and AI/ML by building them*

## EDUCATION

**Carleton University**, Bachelor of Computer Science (Honours) Exp. Graduation: May 2028

- AI & Machine Learning Stream | Minor in Mathematics | **CGPA: 11/12 (A)** | Year 3
- Dean's Honour List | 2026 Clarence C. Gibson Scholarship
- **Coursework:** Systems Programming in C, Operating Systems, Database Management Systems, Software Engineering, Design & Analysis of Algorithms, Linear Algebra, Calculus

**Seneca College**, Electronic Engineering Technician (diploma) May 2019 – May 2021

## TECHNICAL SKILLS

**Languages:** C++, C, Java, Python, SQL, JavaScript, Bash, HTML/CSS
**Frameworks & APIs:** Spring Boot, Spring Security, React, Node.js, REST APIs
**Databases & Cloud:** PostgreSQL, MongoDB, Redis, AWS, Docker
**Tools & AI Development:** Linux, Git, GDB, Make, Maven, Postman, CLion, IntelliJ, GitHub Copilot, Claude Code
**Concepts:** Data Structures & Algorithms, Object-Oriented Design, Design Patterns, CPU Architecture, RISC-V

## PROJECTS

**Database Management System from Scratch** | C++, SQL, JSON, GDB, PostgreSQL

- Built a relational database from scratch in **C++** with a **slotted-page storage engine**, **LRU/CLOCK buffer pool**, SQL parsing, query optimization, and a REPL verified against PostgreSQL
- Processed datasets larger than memory using **external merge sort** and **3 join algorithms** (Block Nested Loop, Sort-Merge, Grace Hash) under a limit as small as 4 pages
- Implemented a **Selinger-style cost-based optimizer** using dynamic programming and histograms to choose join orders for up to 6 tables, validating measured I/O against textbook cost formulas
- Hand-wrote a SQL lexer, **recursive-descent parser**, and semantic analyzer with line/column error diagnostics

**CineSmart: Movie Discovery Platform** | Java, Spring Boot, React, TypeScript, PostgreSQL, Redis, Python

- Built a movie discovery app with a **React/TypeScript** front end and a **Spring Boot REST API** with **36 endpoints**
- Secured authentication with **JWT** access tokens and **rotating refresh tokens** whose reuse detection revokes all sessions, plus **Redis-backed rate limiting** tested for fail-open and fail-closed behavior
- Automated a **Python ETL pipeline** that validates, cleans, and deduplicates data from **3 external movie APIs** into a **16-table PostgreSQL** schema, with daily refreshes of trending titles
- Followed **test-driven development**, writing PostgreSQL and Redis integration tests with **Testcontainers**

**IoT Flood Detection & Alert System** | C, ATmega328P, ESP8266, Embedded Systems

- Developed an **embedded software** system that monitors flood conditions and alerts users remotely in real time
- Wrote **interrupt-driven C firmware** for an **ATmega328P** to read water-level, flow, and humidity sensors live
- Built **ESP8266 Wi-Fi telemetry** that sends live readings to a **cloud dashboard**, with buzzer and phone alerts

## EXPERIENCE

**Carleton University**, IT Support Technician Oct 2026 – Present

- Support **40+ workstations** and an athletics management platform serving **30,000+ users**, diagnosing hardware faults down to the **component level** (RAM, GPU, motherboard, BIOS/CMOS) and resolving **Active Directory**, permissions, and network issues
- **Test and validate** new software updates and application builds in a test environment before department-wide deployment, reporting bugs and usability issues
- Write **end-to-end resolution runbooks** and technical documentation for the department's knowledge base

**CGI**, Technical Support Analyst Jan 2024 – May 2024

- Investigated and resolved **100+ technical tickets** involving complex software and execution errors, using **systematic debugging** to isolate root causes
- Translated technical root causes into clear resolution guidance for non-technical end users